Tuto5: Developing Data Visualizations for multi-dimensional data

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Outline

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Combining D3.js and React: Separation of concerns

- **D3.js visualizations** are implemented in **self-contained class**, without any dependencies to React library
- For each visualization, **one React component** implements the **container of the visualization** and **makes the connection** between the D3js class and the React application.
- The React component:
  - **receives the data** from the Redux store;
  - **instantiates the D3.js class** and call render/update methods to build/update the visualization,
  - **provides the event handler methods to D3.js** class to update the store.

Combining D3.js and React: ready-to-use templates

- `https://github.com/nicolasmedoc/Tuto5-MultiDim-Redux.git`
- Vis-d3.js
  - is a **javascript class** which will renders the visualization. It is self-contained and independent of React
  - declares a method **create()** to initialize the SVG element
  - declares a method **clear()** to remove the SVG from the DOM
  - declares **one or several update methods** (e.g. renderVis(), updateFunction1(), unpdateFunction2()) to change the view or a part of the visualization when the data changes, with the **global update pattern**.

Combining D3.js and React: ready to use templates

- VisContainer.jsx (React component)
  - is the container of the visualization. It uses the React state (useState() hook) or the Redux store (useSelector() hook) and dispatch the actions of reducers to handle the events.
  - controls the component lifecycle with **useEffect() hook** when the component **did mount**, **did update** (when data changes) or **did unmount** (when removed from the page)
  - **instantiates the d3 class** and store it in a **Ref** (with useRef() hook) to keep the d3 instance when the component re-render.
  - **renders the ** element containing the SVG and **stores it in a Ref** (with useRef() hook) to avoid re-render it.

React useEffect() hook to control the component life cycle

3 functions are declared in 2 different profiles of the useEffect() hook to declare additional behavior in the React component life cycle:

```
import { useEffect } from 'react';
...
useEffect(()=>{
    // the behavior after the component creation (did mount)
    return ()=>{
        // the behavior after the component deletion (did unmount)
        // is declared in the return function
    }
}, []) // empty array
useEffect(()=>{
    // behavior after update of dependency1 or dependency2 only
},[dependency1,dependency2]) // array of dependency variables
useEffect(()=>{
    // behavior after update of dependency3 only
},[dependency3]) // array of dependency variables
```

dependency1, **dependency2** and **dependency3** are variables derived from the React state, Redux store (useSelector) or propagated in component props by the parent

React useRef() hook to persist a value in the component

Used to persist the instance object of the D3 class in a React component:

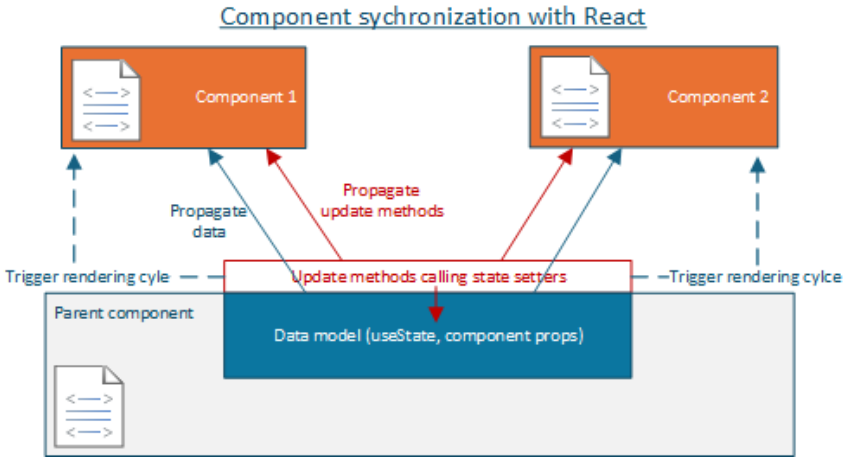
```
import { useEffect, useRef } from 'react';
...
const divContainerRef=useRef(null);
const visD3Ref = useRef(null)
useEffect(()=>{
    const visD3 = new VisD3(divContainerRef.current);
    visD3Ref.current = visD3;
},[])
```

React useRef() hook to persist a value in the component

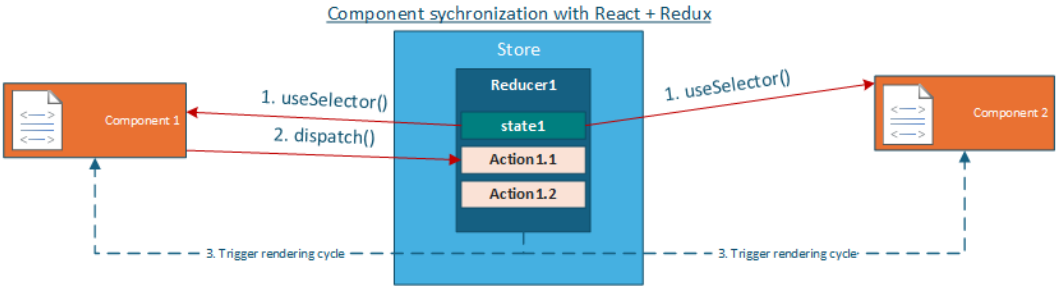
Used to persist the previous value in the sub part of a dataSlice propagated in props:

```
import { useEffect, useRef } from 'react';
...
const dataSliceAttributeRef = useRef(null)
useEffect(()=>{
    // behavior after update of dataSlice
    if(dataSliceAttributeRef.current!==dataSlice.attribute){
        // attribute has been updated => do something
        // e.g. call specific update method in D3 class
        dataSliceAttributeRef.current = dataSlice.attribute
    }
},[dataSlice]) // array of dependency variables
```

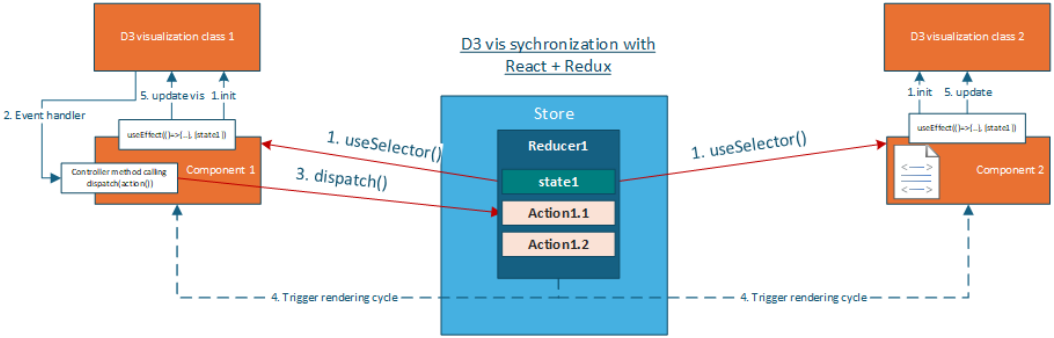
Synchronization of multiple views: React



Synchronization of multiple views: Redux



Synchronization of multiple views: D3 Vis



Building a scatterplot: getting the data set

See in redux/DataSetSlice.js the use of createAsyncThunk() function to retrieve data from data/Housing.csv with async fetch function.
extraReducers in second parameter of createSlice() allows to declare behaviours when the async action is pending, fulfilled or rejected

```
export const dataSetSlice = createSlice({
  ...
  extraReducers: builder => {
    builder.addCase(getDataSet.pending, (state, action) => {
      console.log("extraReducer getDataSet.pending");
      // do something with state, e.g. to change a status
    })
    builder.addCase(getDataSet.fulfilled, (state, action) => {
      return action.payload
    })
    builder.addCase(getDataSet.rejected, (state, action) => {
      // Add any fetched house to the array
      const error = action.payload
      console.log("extraReducer getDataSet.rejected with error" + error);
    })
  }
})
```

Building a scatterplot: getting the data set

Exercise1: In App.jsx, create a useEffect() function to load the dataset when the App component did mount.

Building a scatterplot: render the visualization component

Exercise2: In ScatterplotContainer.jsx call the scatterplotD3.renderScatterplot() method in the useEffect hook reacting to visData updates:

```
// get the current instance of scatterplotD3 from the Ref object...
// call renderScatterplot method ...
// controllerMethods being already defined in the useEffect() method.
// with empty handleOnClick, handleOnMouseEnter and handleOnMouseLeave
```

Building a scatterplot: render the visualization component

Then, add the component in the rendered JSX of App.jsx (x="population" y="ViolentCrimesPerPop")

```
return (
    <div className="App">
        <div id={"MultiviewContainer"} className={"row"}>
         // call the scatterplot component here
        </div>
    </div>
);
```

Building a scatterplot: creation of scales and X/Y axis

In components/scattterplot/Scatterplot-d3.js, see in create() how the xScale and yScale are initialized, and the creation of groups 'g' for x and y axes.

Exercise3: in updateAxis(), using d3.min(mylist) and d3.max(mylist), set the domain values of this.xScale.domain(...) and this.yScale.domain(...) to put **"area" in X Axis and "price" in Y axis**.

In "updateAxis" function, use these scales to build X axis and Y axis:

```
// .xAxisG and .yAxisG are initialized in create() function
this.svg.select(".xAxisG")
    .transition().duration(500)
    .call(d3.axisBottom(this.xScale))
;
this.svg.select(".yAxisG")
    .transition().duration(500)
    .call(d3.axisLeft(this.yScale))
;
```

Building a scatterplot: update circle positions with scales

in components/scattterplot/Scatterplot-d3.js, in the method updateMarkers():
Exercise4: using X/Y scales, apply a translation to .markerG to update the circle positions. updateMarkers(selection) is called from renderScatterplot(). The "selection" parameter comes from the update pattern using join(). It corresponds to a selection of ".markerG".

```
selection
    .transition().duration(500)
    .attr("transform", (itemData)=>{
        // use scales to return shape position from data values
})
```

Building a scatterplot: reuse the scatterplot component

All results of previous exercises are in the branch "scat" **Exercise 5**: display two scatterplot side by side to show different pair of dimensions:

- scatterplot 1: (x="population" y="ViolentCrimesPerPop")
- scatterplot 2: (x="medIncome" y="ViolentCrimesPerPop")

Coordinated multiple views: principles

- **Exercise 6:** synchronize the two scatterplots on click interaction. The purpose is to highlight the clicked objects simultaneously in the two scatterplots (e.g. make the red border visible).

- **In ScatterplotContainer.jsx useEffect()**: the object controllerMethods is propagated to the ScatterplotD3 class and contains all the methods defining the behaviour of interactions (handleOnClick, handleOnMouseEnter, handleOnMouseLeave)

- These methods have to **update a data slice in the redux store** to synchonize the two scatterplots by calling a dispatch with the corresponding reducer actions.

Coordinated multiple views: prepare the actions in the reducer

Create a new redux slice (ItemInteractionSlice.js) that will contain an object with an array selectedItems and an object hoveredItem selectedItems:[], hoveredItem:. Create the reducer actions setSelectedItems and setHoveredItem. Don't forget to declare the reducer in the store.

Coordinated multiple views: use dispatch to call the action functions

In ScatterplotContainer.jsx call the reducer action with dispatch in handleOnClick, which is propagated to the D3 class.

Coordinated multiple views: highlight an object in the views

Propagate the synchronization data in ScatterplotContainer.jsx:

- with useSelector(), retrieve selectedItems
- create a specific useEffect with selectedItems as dependency to call the function in D3 class that will highlight the selected marker: highlightSelectedItems()

Coordinated multiple views: highlight an object in the views

In Scatterplot-d3.js highlightSelectedItems method, use the update pattern to declare the right behaviour:

```
highlightSelectedItems(selectedItems){
    // use pattern update to change the border and opacity of objects:
    //      - call this.changeBorderAndOpacity(selection,true)
    //        for markers that match selectedItems
    //      - this.changeBorderAndOpacity(selection,false) for
    //        markers that do not match selectedItems
}
```